

AMR SAEED

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SUMMARY

Results-oriented Computer and Systems Engineering developer combining a strong software foundation with solid hardware logic. Specialized in cross-platform mobile development (Flutter, React Native) and structured custom backend architectures (Node.js, MongoDB Atlas). Strong background in Embedded Systems, IoT protocols, and RTOS design. Highly motivated to write Clean Architecture code, apply robust state management, and deliver scalable, high-performance software engineering solutions.

KEY PROJECTS

Loadr | Universal Media Engine (Flutter, Python, FFmpeg) | [GitHub Repo](#)

- Built a client-side media extraction engine running an embedded native Python environment (yt-dlp) via Flutter Platform Channels.
- Bypassed centralized scraping servers and IP rate limits by routing stream requests directly through local residential networks.

Batrina | Full-Stack Social E-Commerce Platform | [GitHub Repo](#)

- Developed a full-stack social e-commerce platform architected around strict Clean Architecture principles and MVVM modular design.
- Built a custom high-performance Node.js & MongoDB REST API with low-latency bidirectional notifications via Firebase Cloud Messaging (FCM).

Portfolio | High-Performance Interactive Web Showcase | [GitHub Repo](#)

- Built an interactive GPU-accelerated developer portfolio using React 18, Vite, GSAP ScrollTrigger, Framer Motion, and pure Vanilla CSS3 & Flexbox.
- Engineered a custom 60FPS Bezier-trajectory Particle Canvas engine with dynamic real-time DOM element bounding box sampling.

Green Guardian | IoT Smart Plant Monitor (ESP32, RTOS, Flutter) | [GitHub Repo](#)

- Engineered an agricultural IoT telemetry system for real-time plant health monitoring and automated irrigation utilizing an ESP32 and FreeRTOS.
- Built a high-responsiveness companion Flutter dashboard app for live diagnostic visualization and remote actuator control via WiFi/Blynk.

MIPS-32 CPU Design | VHDL, Quartus Prime, FPGA | [GitHub Repo](#)

- Designed and modeled a fully functional 32-bit pipelined RISC MIPS processor architecture using synthesized VHDL to master hardware logic.
- Implemented 5 complete instruction execution stages (IF, ID, EX, MEM, WB) with algorithmic data hazard detection and forwarding mechanisms.

EXPERIENCE & INTERNSHIPS

Mobile Application Trainee, DEPI (Ministry of Communications)	2025
- Selected for an intensive, government-sponsored training program focused on practical mobile development utilizing Flutter. - Performed hands-on tasks building dynamic UI components, applying reactive state management, and seamlessly integrating RESTful APIs.	
Mobile Application Trainee, Information Technology Institute (ITI)	2024
- Received comprehensive training on fundamental and advanced Flutter concepts. - Built structured app prototypes focusing on clean architecture principles, efficient state management, and modular code design.	

EDUCATION

Bachelor of Science in Computer and Systems Engineering	Sep 2021 - Present.
Zagazig University, Egypt	
Core Coursework: Embedded Systems, Software Engineering, Data Structures, Microprocessors, and Clean Architecture.	

TECHNICAL SKILLS

Mobile Development: Flutter, React Native, Dart, Clean Architecture, MVVM.
Web & Backend: Node.js, Express, MongoDB Atlas, React, Vanilla CSS3, REST.
Hardware & IoT: C/C++, RTOS (FreeRTOS), VHDL/FPGA, ESP32, Arduino, Embedded.
Tools & Ecosystem: Git, GitHub, Postman, Figma, Vercel, Linux CLI, CI/CD.

LANGUAGES

- Languages: Arabic (Native)
- English (Very Good)